

The World Model as Judge: Learned Traversal Rewards for Humanoid Navigation in Cluttered, Human-Occupied Spaces

(working notes retained in *paper/planning.md*; this file converges toward the submission text.
Target venues: ICRA / ICLR.)

Abstract

Humanoid robots can walk, crawl, and recover from pushes, yet they still cannot traverse the narrow, cluttered, human-occupied spaces where they are expected to work. The bottleneck is not low-level control but **reward specification**: behaviors such as turning through a tight gap, crouching under furniture, or yielding to a person are inherently perceptual and resist hand-crafted reward engineering. We propose using a video-language world model as a learned reward model. We fine-tune a compact 2B-parameter model into a traversal critic that assigns ordinal quality scores to short navigation clips, supervised entirely by privileged simulator state converted automatically into labels — no human annotation is required. The critic provides reward shaping for a lightweight vision-based policy that commands a frozen whole-body controller, and it is used at training time only: no world model is deployed on the robot. On scenes held out at both the layout and appearance level, fine-tuning raises the critic's correlation with ground-truth traversal quality from $r = 0.02$ to $r = 0.48$, scaling with data. [P2 closing sentence: "We further show that critic-shaped policies collide less often than hand-crafted-reward baselines at matched success rates" – swap in measured numbers, or if P2 slips: "Finally, we describe an asynchronous scoring architecture in which the critic never blocks policy optimization."]

1. Introduction

1.1 Introduction (opening)

Humanoid locomotion has advanced rapidly: whole-body controllers trained with large-scale simulation RL now walk, crawl, and recover from pushes, and behavior foundation models such as GEAR-SONIC compress thousands of hours of human motion into a single policy. What remains unsolved is where those robots are supposed to go: through the 0.6-meter gap between the sofa and the bookshelf, under the low table, past the person carrying groceries. This is not a control problem — the controller already knows how to crouch — it is a **decision and reward problem**. No practical reward function captures "traverse this room as a considerate person would"; hand-crafted proxies — clearance penalties, velocity bonuses, personal-space costs — are brittle surrogates that RL readily exploits. We show that a compact video-language model, fine-tuned on automatically labeled simulation clips, can predict a privileged-state traversal-quality score from pixels alone on held-out scenes ($r = 0.48$) — a prerequisite for using it as a learned shaping reward. Because it evaluates quality from pixels rather than privileged state, it can in principle be applied where privileged state

does not exist; we present preliminary positive-transfer evidence on real robot footage and identify viewpoint sensitivity as an open confound.

1.2 The observation the paper is built on

Generative video world models are used two ways in robotics today: as **simulators** (roll out imagined futures for planning) and as **policies** (action-conditioned VLAs). Both uses stress exactly what these models are worst at — long-horizon pixel-consistent generation — and both put a large model in or near the control loop. We exploit the third option: a video model as a **judge**. Recognition is easier than generation; scoring "was that traversal safe and natural?" needs one VLM forward pass over a short clip, not thirty diffusion steps per imagined frame. Judging is also the only use that is **training-time only**: the deployed system is a ~ 0.5 M-parameter policy on top of a frozen controller, with no world model on the robot.

1.3 Contributions

1. **A method**: fine-tune a compact video-language world model into a traversal critic — an ordinal (1-5) scorer of navigation clips — supervised entirely by privileged simulator state via a deterministic auto-labeler (four interpretable axes: safety, clearance, social, progress; safety as hard ceiling). No human labels; supervision scales with simulation.
2. **Evidence the premise holds**: on scenes held out at the scene level (novel layouts, gap widths, furniture, appearance), the critic recovers ground-truth traversal quality from pixels alone: Pearson $r = 0.02$ (pretrained, zero-shot) $\rightarrow 0.38$ (460 training clips) $\rightarrow \mathbf{0.48}$ (1,530 clips), $\text{acc} \pm 1 \ 0.755$ — a clean monotone scaling curve. [v3: balanced $\sim 2,900$ -clip result]
3. **A system**: a three-layer architecture (frozen SONIC whole-body controller at 50 Hz; small vision policy at ~ 3 Hz; critic as asynchronous training-time reward) with an async scoring protocol that never blocks RL on the 2B model. [P2: paired-seed comparison – critic-shaped vs. hand-crafted-reward PPO on held-out scenes.]
4. **A quantified failure case for hand-crafted rewards**: our initial baseline reward silently collapsed the policy to standing still — per-frame contact penalties dominated the progress term — illustrating the brittleness the critic is designed to remove.

1.4 Scope and assumptions

We do not claim the critic replaces task reward: sparse goal reward and privileged safety terms remain the backbone; the critic shapes. We do not claim video world models should replace simulators. And the current experiments are in simulation with kinematic playback of a pretrained motion planner; physics-in-the-loop training and real-robot deployment are staged as follow-up (§7).

2. Related Work

(bullets to be expanded to cited prose; mandatory additions per review: VLM-RMs (Rocamonde et al. 2023), RL-VLM-F (Wang et al. 2024), Eureka (Ma et al. 2023), VIPER (Escontrela et al. 2023), RoboCLIP, MineCLIP, Christiano et al. 2017/PEBBLE, VIP/LIV/R3M)

- **Humanoid whole-body control / behavior foundation models.** GEAR-SONIC, Decoupled WBC (GR00T), BeyondMimic, unitree_rl_lab. We consume these frozen.
- **Cluttered-space humanoid traversal.** HumanoidPF / Click-and-Traverse (G1, MuJoCo, procedural obstacles with difficulty knobs) — closest prior work; static clutter, hand-crafted rewards. Our deltas: dynamic people, learned perceptual reward, style-mode action space.
- **Social navigation.** Classic (ORCA, social forces), learned (NavIsaacLab crowds, Habitat 3.0 social nav). Reward is invariably hand-crafted; our critic is the missing learned-reward piece.
- **World models in robotics.** As simulators (dreamer-style, video prediction for MPC), as policies (VLA: GR00T N-series, π_0 , Cosmos policy mode). As judges: VideoPhy-2-style physical-plausibility scoring is the nearest neighbor — we extend the idea from "is this video physical?" to "is this traversal good?" and close the loop into RL.
- **Learned rewards / RLHF / VLM-as-reward.** Reward models from preferences, VLM zero-shot rewards (e.g., CLIP-based), foundation-model feedback for RL. Our distinction: video (not image) judgment, ordinal calibrated output, auto-labels from privileged sim state rather than human preference.

3. Method

3.1 Problem setup

Goal-conditioned navigation in cluttered indoor scenes with moving people. Deployed stack: frozen whole-body controller **C** (SONIC: latent-token motion tracking at 50 Hz + kinematic planner exposing {mode, direction, speed, height}); small vision policy **π** (egocentric RGB + goal vector → planner commands at ~3 Hz). Train π with PPO; the research question is the reward.

3.2 Traversal critic

Data. Parameterized procedural scenes (doorway gaps 0.6–1.2 m, furniture, low tables, moving people on crossing paths; per-scene visual randomization). Scripted policies spanning five quality bands generate diverse rollouts. Privileged channels recorded per frame: min clearance (analytic point-to-box), min person distance, base speed, goal distance, contact flags.

Auto-labeler. Four axis scores from documented thresholds — safety (contact duration), clearance (5th-percentile margin + slow-when-tight bonus), social (min person distance), progress (goal-rate + freeze penalty) — composed as overall = min(safety, half-down-round(mean)). Safety is a hard ceiling: a collision episode cannot score well on style. Scene-hash split: validation scenes are entirely unseen (layout and appearance).

Model. Cosmos3-Edge reasoner (Nemotron-2B LM + SigLIP2 tower), SFT on (clip, rubric prompt) → single digit; partial unfreeze (projector + last 8 LM blocks + final norm) fits a single 22 GiB L4. Class-balanced manifest oversampling corrects ordinal imbalance. [v3]

Two implementation details are essential for reproduction: (i) the chat template's generation-time reasoning mode must be disabled for a critic fine-tuned to emit a single digit; otherwise every checkpoint produces free-form chain-of-thought and no parseable score. (ii) Validation loss is the wrong model-selection signal for this ordinal scorer — it reached its minimum at iteration 200 while downstream correlation kept improving through iteration 400; select checkpoints on task metrics.

3.3 Critic-shaped RL

$R = R_{\text{task}}(\text{sparse goal} + \text{dense progress}) + R_{\text{safety}}(\text{privileged, bounded})$

- $\lambda \cdot (\text{critic}(\text{clip}) - 3)/2$, λ bounded. The critic runs **asynchronously**: the env banks episode clips to a file queue; a scorer daemon (owning the GPU) returns scores; bonuses fold into later batches. RL never blocks on the 2B model. Anti-hacking: sparse+privileged terms remain the backbone; periodic audit of top-decile-critic episodes against the auto-labeler.

4. Experiments

4.1 Can the critic read traversal quality from pixels?

Held-out-scene evaluation, ordinal metrics (exact acc, $\text{acc} \pm 1$, Pearson, Spearman, per-class recall).

Model	train clips	val clips	Pearson	$\text{acc} \pm 1$	acc
near-base (20 clips seen)	—	140	0.024	0.557	0.314
critic v1 (best ckpt)	460	140	0.377	0.679	0.457
critic v2 (best ckpt)	1,530	470	0.482	0.755	0.517
critic v3 (balanced)	2,930	670	0.452	0.743	0.460
critic v4 (clean camera)	2,410	590	0.534	0.768	0.476

Scaling is monotone in data and steps through v2. **v3 isolates class balancing** (manifest oversampling to uniform per-score weight): overall correlation holds $\approx$ flat on a harder, larger val while tail-class recall roughly doubles — GT-1 0.22→0.35, GT-5 0.06→0.15, GT-3 0.12→0.24 (Fig. 5). That is the right trade for a reward model: mistaking a dangerous episode for a good one is the costly error, and it is precisely the rare-class confusions that balancing removes. **v4 isolates the camera confound** (§4.4 note): v1–v3 clips blank out the robot for 30–47% of frames in some regimes (chase camera below wall height; wall-fill fraction correlated with score band); v4 regenerates all data with an above-wall view. The clean-camera critic is the best to date — Pearson 0.534 [0.469, 0.594], $\text{acc} \pm 1$ 0.768 [0.732, 0.803] (95% clip-bootstrap CIs; best checkpoint iter 700 of 800) — and its confusion matrix is substantially rebalanced: GT-5 recall 0.15 → 0.42, GT-1 0.35 → 0.47, GT-3 0.19 → 0.35 relative

to v3, at similar overall accuracy. A wall-fill-only shortcut probe explains $r = 0.155$, confirming occlusion statistics were not the critic's signal.

4.2 Does critic shaping improve the policy? [P2 – headline]

Paired-seed PPO, identical scenes/architecture/steps; arms: (a) sparse+safety baseline, (b) + hand-crafted dense shaping, (c) + critic shaping. Endpoint metrics on held-out scenes: success rate, collision-episode rate, min clearance, person-space violations, time-to-goal, and auto-labeler axis scores (one ruler for policies and critic).

Baseline arm (measured). 300k-step PPO with hand-crafted reward (progress + bounded contact + time). Training-scene success peaked ~ 0.43 around 112k steps and degraded to ~ 0.32 by 300k (instability without KL/trust constraints on a small net). On the 40-episode held-out-scene eval:

checkpoint	success	collision-ep rate	mean auto-labeler score
baseline @ peak (112k)	0.15	1.00	1.93
baseline @ last (300k)	0.10	0.98	1.83

The train-to-held-out gap ($0.43 \rightarrow 0.15$) and the near-100% collision rate quantify the headroom the critic-shaped arm targets. Notably, the hand-crafted reward reached moderate training-scene success, yet its policy collides in nearly every held-out episode and scores below 2 on the auto-labeler scale.

4.3 Transfer to real robot video

We score 42 clips of **real Unitree G1 footage** (GEAR-SONIC release media: in-the-wild navigation, style walks, impaired gait, crawling, kneeling) plus 27 clips from a different simulator, using the best sim-trained critic. No privileged state exists for this footage, so no auto-label can.

Result (v4 best checkpoint, decode path verified to match training): **zero parse failures; every clip scored in the 4-5 band** — the correct range, since all release demos are collision-free. Clean walking scores highest among multi-clip real regimes (mean 4.80, 8/10 clips at 5), crawling lowest (4.0), with posture-change and the different-simulator control in between (4.5 / 4.63). The critic transfers to real video without collapsing to noise or misreading clean demos as unsafe. (The single impaired-gait clip scored 5 — $n=1$; the auto-labeler's rubric has no gait-quality channel, so this is consistent with its supervision, and a reminder that the critic inherits its labeler's blind spots.) Two limitations qualify this result: (i) the released footage contains no negative examples (no real collisions), so this test establishes transfer without collapse, not full discrimination on real video — closing that requires collecting deliberately-flawed real rollouts; (ii) score compression toward 4 mirrors the training distribution's mode.

4.4 Further ablations (planned)

Critic checkpoint quality vs. policy gain; λ sensitivity; band-3 "clean but sloppy" discrimination; real negative-example collection.

4.5 Failure modes of hand-crafted rewards

First baseline reward: success $0.18 \rightarrow 0.00$ over 150k steps as PPO discovered that standing still dominates walking (per-frame contact penalties $\times 15$ frames/step $>$ max progress term). A single line of reward arithmetic caused a silent collapse. This is precisely the failure mode a learned critic avoids: its judgment is holistic and episode-level rather than a sum of per-frame proxies that the policy can exploit.

5. Figures

- **Fig 1 (system)**: three-layer architecture; what deploys vs. what judges. $\rightarrow$ [paper/figures/fig1_system.svg](#)
- **Fig 2 (scaling)**: critic Pearson vs. training clips ($0.02 \rightarrow 0.38 \rightarrow 0.48 \rightarrow [v3]$), with $\text{acc} \pm 1$ as secondary panel. $\rightarrow$ [paper/figures/fig2_scaling.html](#) (export to PDF/SVG for submission)
- **Fig 3 (qualitative)**: five clips, one per score, with critic score vs. auto-labeler score; egocentric frames strip.
- **Fig 4 (P2 headline)**: paired bars — the three PPO arms on the endpoint metrics.
- **Fig 5 (confusion)**: v2 vs. v3 confusion matrices — the class-balance ablation, visually.

Rendered figures

Fig. 1 — The world model as judge: the critic shapes training-time reward and never deploys.

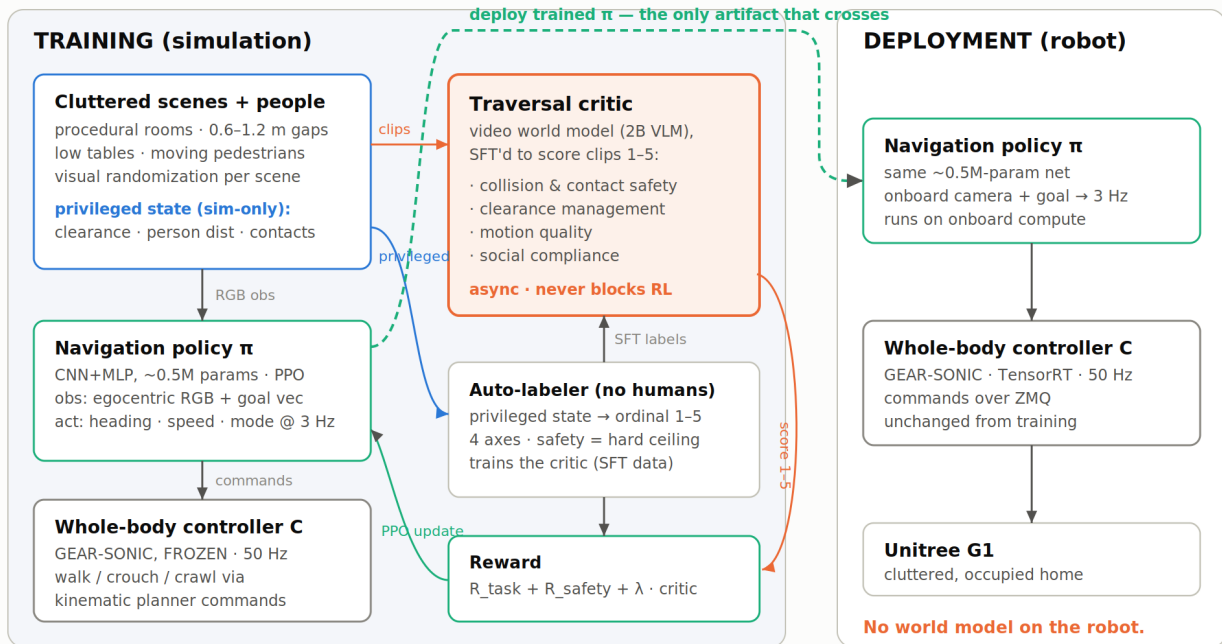


Fig. 2 — Critic quality vs. training data (scene-level holdout, 95% clip-bootstrap CIs)

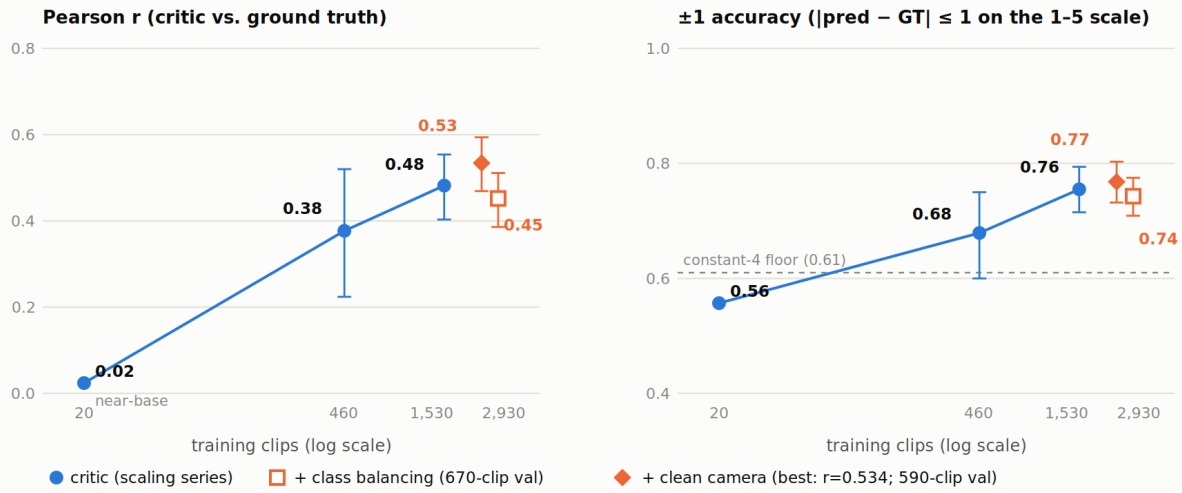


Fig. 3 — One episode per ground-truth score (chase camera; time left to right)

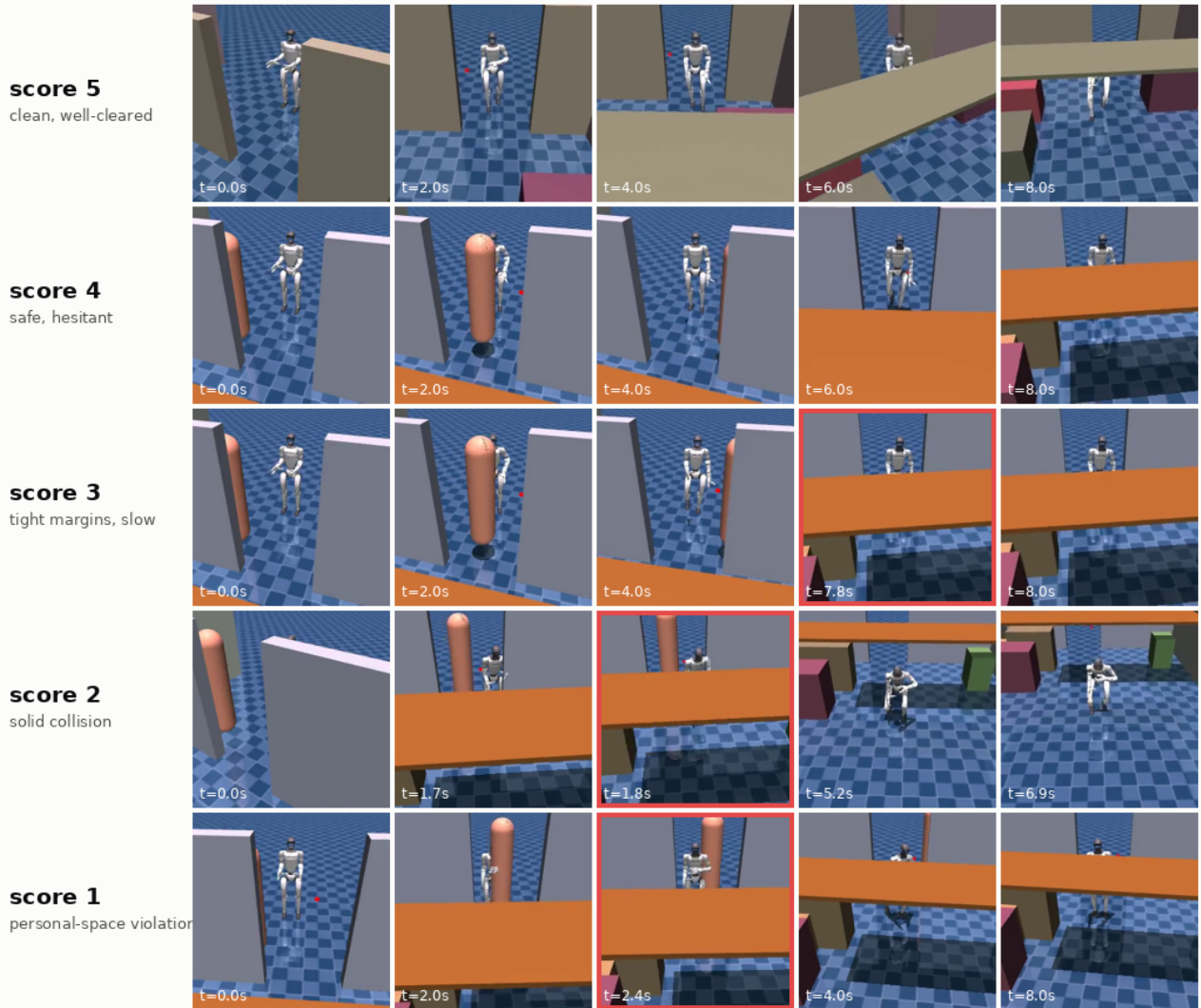


Fig. 5 — Class balancing shifts recall toward the rare scores (row-normalized; n per row at right)

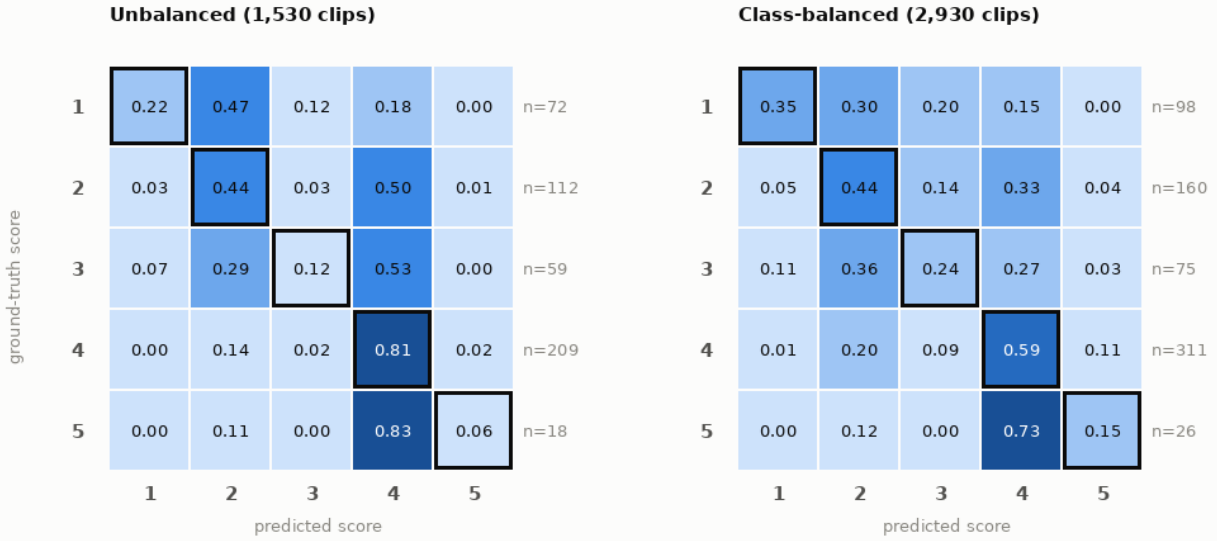
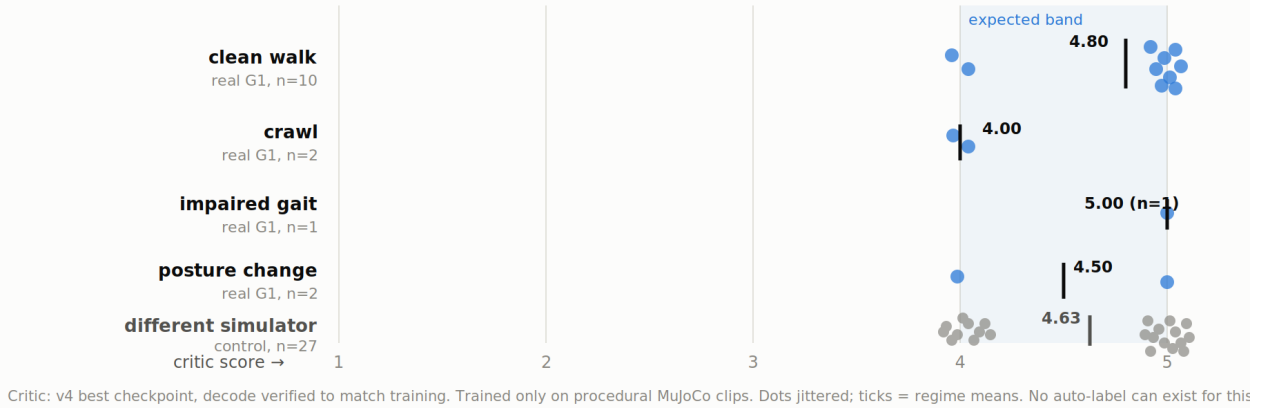


Fig. 6 — The sim-trained critic transfers to real robot video

42 real G1 clips + 27 other-sim controls · zero parse failures · regime ordering emerges: clean walk 4.80 > crawl 4.00



6. Reproducibility notes

Single shared 22 GiB L4 for everything (critic SFT, eval, scoring); CPU for rollout generation (planner ONNX ~35 ms) and PPO. Full recipe: branch `feat/traversal-critic` — generator, labeler (+18 unit tests), SFT SKU/TOML, eval sweep, scorer daemon, nav env, PPO trainer, paired evaluator.

7. Limitations & staging

Kinematic playback (no physics response) in phase 1-2; SONIC tracker physics-in-the-loop is phase 2-proper (Isaac Lab, pinned 4.5, GRScenes-100 + people layer); real-G1 deployment is phase 4 (ZMQ command path verified in repo docs). Critic gate (Pearson ≥ 0.7) not yet met — scaling curve suggests data, not architecture, is the binding constraint. Auto-labels inherit threshold choices; the human-label override path exists for calibration.